

Pythia-160M Repeat-Match Alignment Trajectory

Cross-seed causal transfer by raw-score alignment versus same head index

Branch specialization research project

2026-05-22

Checkpoint reason: this is the first Pythia-160M training-trajectory test of whether a causal head role transfers across seeds by raw index or by learned raw-score alignment.

Question

The 14M trajectory showed that repeat-match attention specialization can appear before strong causal head importance.

The stronger 160M question is:

when the role becomes causal, is it in the same head slot across seeds, or in an aligned but relabeled slot?

This separates strong universality from weak, relabeled role universality.

- Models: EleutherAI/pythia-160m-seed1, seed2, seed3
- Revisions: step0, 1000, 4000, 16000, 64000, 143000
- Probe: repeated-token sequences $[x_1, \dots, x_{32}, x_1, \dots, x_{32}]$
- Selected heads: top repeat-match head in layers 0 and 1
- Alignment: checkpoint-specific Hungarian matching on raw attention scores from four fixed natural probe texts
- Causal test: zero selected head outputs, then measure second-half continuation loss

Metric Glossary

Probe specialization Normalized repeat-match attention score of the selected heads. It measures an attention pattern, not causality.

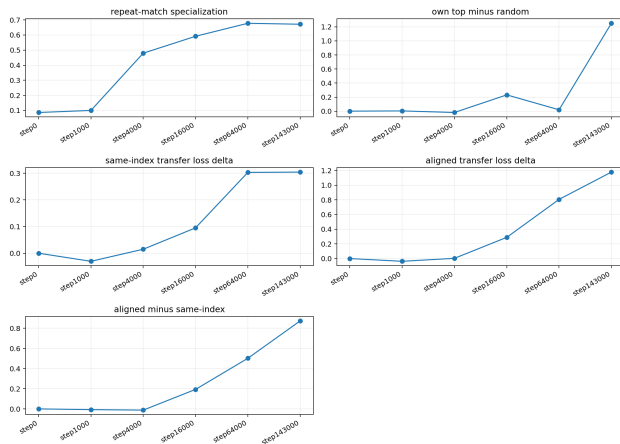
Own top minus random Own selected-head loss delta minus random same-layer controls. It asks whether the probed heads matter causally.

Same-index transfer Ablate the target seed's heads at the source seed's original layer/head indices.

Aligned transfer Ablate target heads matched to source heads by raw-score Hungarian alignment.

Aligned minus same Paired advantage of aligned transfer over same-index transfer.

Trajectory



Probe specialization rises by step4000, but aligned causal transfer separates from same-index transfer mainly from step16000 onward.

Aggregate Results

Revision	Probe spec.	Own top – rand.	Same-index	Aligned	Aligned – same
step0	0.0859	0.0011	-0.0000	0.0005	0.0005
step1000	0.0995	0.0042	-0.0304	-0.0371	-0.0067
step4000	0.4794	-0.0180	0.0149	0.0034	-0.0114
step16000	0.5916	0.2324	0.0952	0.2890	0.1939
step64000	0.6778	0.0206	0.3034	0.8057	0.5023
step143000	0.6715	1.2500	0.3046	1.1774	0.8728

Means over seeds 1-3. Transfer means use six ordered source-target seed pairs per checkpoint.

Paired Transfer Test

Revision	Pairs	Aligned better	Aligned – same mean
step0	6	4	0.0005
step1000	6	2	-0.0067
step4000	6	4	-0.0114
step16000	6	5	0.1939
step64000	6	5	0.5023
step143000	6	6	0.8728

By the final checkpoint, raw-score alignment beats same-index transfer in all six ordered seed pairs.

Supported

- Probe specialization appears early: selected-head specialization is already 0.4794 by step4000.
- Causal importance appears later: own top minus random reaches 0.2324 at step16000 and 1.2500 at step143000.
- Cross-seed causal role identity is mostly relabeled: final aligned transfer is 1.1774 versus 0.3046 for same-index transfer.

Not shown

- This does not prove functional modularity in ordinary Pythia heads.
- It validates alignment as a causal role-transfer metric for this repeated-token behavior.

Caveats

- Three seeds only, not the full nine-seed Pythia suite.
- Layers 0 and 1 only.
- Synthetic repeated-token task.
- Alignment uses four natural probe texts; this should be expanded.
- Head-output ablation, not full path patching.
- Random controls are noisy at step64000, so aligned-vs-same transfer is the cleaner cross-seed readout there.

Project-Level Update

The Phase 1 story is now stronger:

- ① Same head index is a weak carrier of the repeat-match role across seeds.
- ② Raw-score alignment reveals the transferable causal role after training has progressed far enough.
- ③ Probe-defined specialization and causal transfer have different developmental timelines.

The paper framing should distinguish role existence, causal importance, and alignment-based cross-seed identity.

Decision And Next Step

Decision. Keep raw-score alignment plus causal transfer as the main Phase 1 test of weak universality.

Next step. Scale the 160M result before making paper-level claims:

- run seeds 1-9 at step4000, step16000, and step143000;
- expand the alignment probe corpus;
- keep same-index transfer as the negative baseline;
- optionally add path patching after the ablation result survives.

Provenance

Source

- `scripts/pythia_repeat_match_alignment_trajectory.py`
- `scripts/analyze_pythia_alignment_trajectory.py`
- `doc/phase1_pythia160m_repeat_match_alignment_trajectory.md`

Copied data

- `data/revision_summary.csv`
- `data/condition_summary.csv`
- `data/transfer_pair_summary.csv`
- `figures/repeat_match_alignment_trajectory.png`

Run command

- `python3 -u scripts/pythia_repeat_match_alignment_trajectory.py --model-size 160m --seeds 1 2 3 ...`